DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-100-02 — Bench test relief valve 3"-P150-RV-4002/2 on T-101-N3

Area	A100 Tank farm	**Task**	PR Relief valve bench test and reinstatement
Priority	1 (safety)	**Issued by**	Facility owner (demonstration)
Primary asset	**T-101** — T-101 tank 12' dia × 22' high — E 25'-0" N 26'-6" EL 100'-6"		
Equipment	T-101	**Lines**	3"-P150-RV-4002, 4"-P150-FL-4005
Isometrics	ISO-3"-P150-RV-4002, ISO-4"-P150-FL-4005 (generated); owner's drawings ISO-RV-4002, ISO-FL-4005		
Permits	General work permit, Lifting plan		

Reason

Relief valve due for its periodic bench test.

Scope of work

1. Confirm T-101 depressured and isolated.
2. Unbolt and remove VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OUTLET (3"-P150-RV-4002/2); blind the nozzle and the outlet line.
3. Bench test: set pressure 150 psi and seat tightness (SW-TEST-3); service the valve as found.
4. Reinstall with new gaskets and studs, tightened per SW-WO-5; remove blinds.

Affected items: 3"-P150-RV-4002/2, 3"-P150-RV-4002/1, 4"-P150-FL-4005/A1

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-3, SW-WO-1, SW-WO-2, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-RV-4002	3"-P150-RV-4002	3"-P150-RV-4002	1	0 error(s), 1 warning(s) — see below
ISO-FL-4005	4"-P150-FL-4005	4"-P150-FL-4005	3	0 error(s), 2 warning(s) — see below

ISO-RV-4002 (PDF) · ISO-FL-4005 (PDF) — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-RV-4002		warning	TO label 'PSV' is not on the drawing
ISO-FL-4005	3	warning	weld 4"-P150-FL-4005/W1 (E 22'-6" N 4'-6" EL 123'-6 11/16", sheet 3) has no W1 label
ISO-FL-4005		warning	FROM label 'PSV ON T-101-N3' is not on the drawing

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
T-101	equipment	T-101 tank 12' dia × 22' high		E 25'-0" N 26'-6" EL 100'-6"
3"-P150-RV-4002/2	item	VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OUTLET	3"-P150-RV-4002	E 22'-0" N 26'-6" EL 124'-0"
3"-P150-RV-4002/1	item	GASKET 3" P150 COMPRESSED SHEET	3"-P150-RV-4002	E 22'-0" N 26'-6" EL 123'-4-3/8"
4"-P150-FL-4005/A1	item	GASKET 4" P150 COMPRESSED SHEET	4"-P150-FL-4005	E 22'-0" N 25'-10-1/16" EL 124'-0"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	T-101	3"-P150-RV-4002	T-101 depressured, drained and isolated under the shutdown isolation (no inlet valve)			
2	blind	3"-P150-RV-4002/1	3"-P150-RV-4002	Blind the nozzle T-101-N3 once the valve is removed	E 22'-0" N 26'-6" EL 123'-4-5/16"		
3	blind	4"-P150-FL-4005/A1	4"-P150-FL-4005	Blind the outlet line 4"-P150-FL-4005 at its first flange	E 22'-0" N 25'-10-1/8" EL 124'-0"		

Parts

Description	Qty	Replaces
GASKET 3" P150 COMPRESSED SHEET	1	3"-P150-RV-4002/1
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P150	1	3"-P150-RV-4002/3
GASKET 4" P150 COMPRESSED SHEET	1	4"-P150-FL-4005/A1
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	1	4"-P150-FL-4005/A3

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
3"-P150-RV-4002/1	4 × 0.875" studs (3"-P150-RV-4002/3)	27,059	3	
4"-P150-FL-4005/A1	4 × 1" studs (4"-P150-FL-4005/A3)	35,343	3	

Test — Relief valve bench test

Description	Set pressure 150 psi (class design pressure, PSV-1) and seat tightness
Pressure	150 psi
Basis	SW-TEST-3

Inspection and test plan (SW-ITP-1..3)

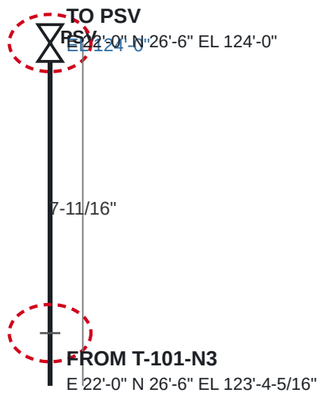
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Relief valve removed; nozzle and outlet blinded	SW-WO-3	Both openings blinded and tagged	Isolation certificate	W	W	—				
3	As-found bench test (pop pressure recorded before any service)	SW-TEST-3	Recorded	Bench test record	W	W	W				
4	Set pressure and seat tightness after service	SW-TEST-3, PSV-1	Set at 150 psi; seat tight	Bench test record, seal tag	H	H	W				
5	Valve reinstalled; new gaskets; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Isometric ISO-3"-P150-RV-4002

ISOMETRIC — 3"-P150-RV-4002 — Bench test relief valve 3"-P150-RV-4002/2 on T-101-N3

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BILL OF MATERIALS

1	GASKET 3" P150 COMPRESSED SHEET	1
2	VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OU…	1
3	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P150	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3"-P150-RV-4002

CLASS H150 — Hydrocarbon process
SERVICE RV · 3" · P150 · DESIGN 150 psi
T-101-N3 → PSV
PIPE 0.000" · WEIGHT 114 lb

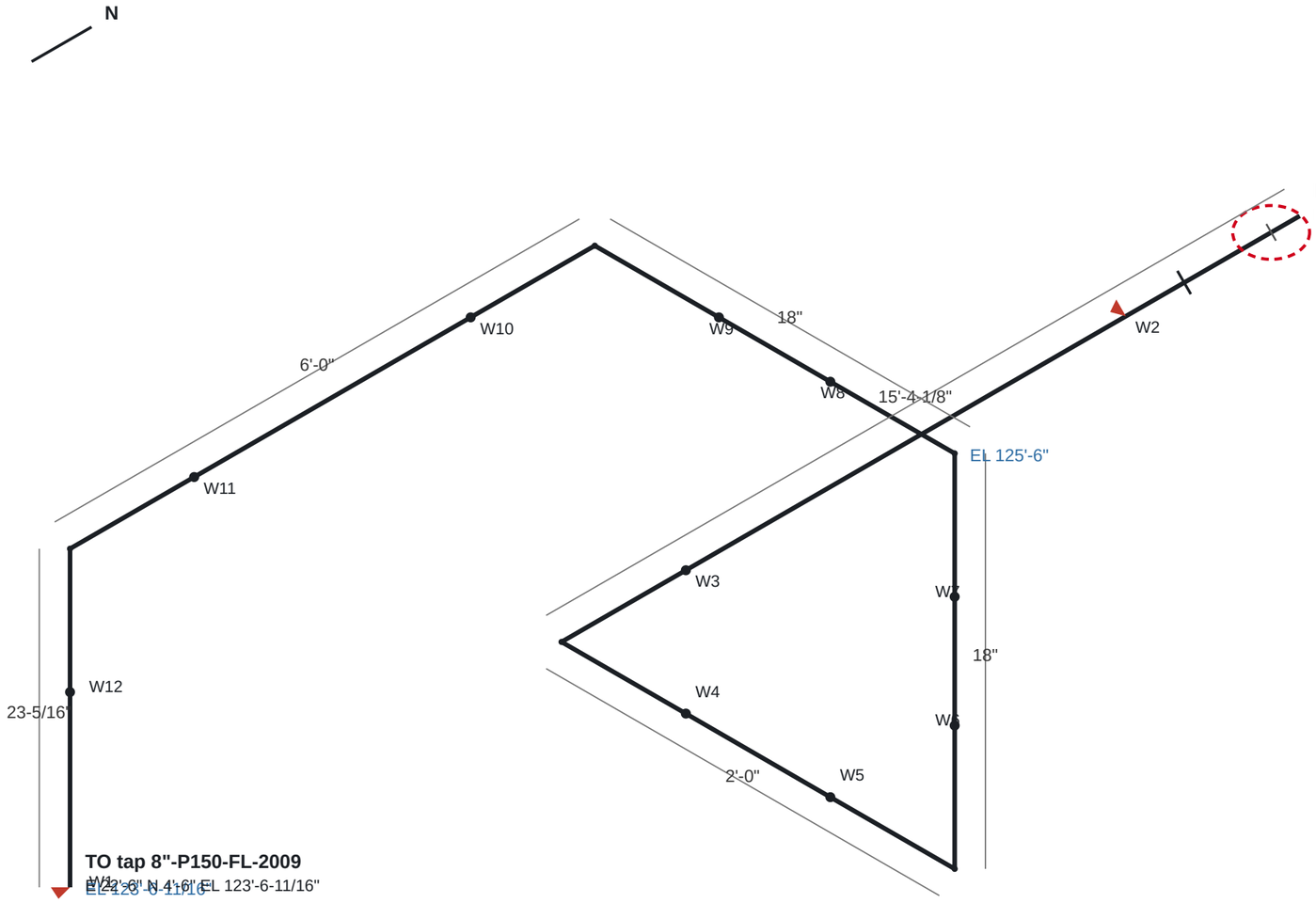
DWG ISO-3"-P150-RV-4002 REV 0 WO-100-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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Isometric ISO-4"-P150-FL-4005

ISOMETRIC — 4"-P150-FL-4005 — Bench test relief valve 3"-P150-RV-4002/2 on T-101-N3



FROM PSV on T-101-N3
E 22'-0" N 25'-10-1/8" EL 124'-0"

BILL OF MATERIALS

1	GASKET 4" P150 COMPRESSED SHEET	1
2	FLANGE WELD-NECK 4" P150 RAISED FACE	1
3	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	1
4	PIPE 4" P150 BEVELLED ENDS	22'-3-1/4"
5	ELBOW 90° R = 5/3 OD 4" P150	5

WELDS 12 (2 FIELD) · SUPPORTS 0 · SPOOLS 2

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11 W12

4"-P150-FL-4005

CLASS F150 — Flare and relief outlets
SERVICE FL · 4" · P150 · DESIGN 150 psi
PSV on T-101-N3 → tap 8"-P150-FL-2009
PIPE 22'-3-1/4" · WEIGHT 240 lb

DWG ISO-4"-P150-FL-4005 REV 0 WO-100-02

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